ExamVision: Classifying Student Attention States During Remote Examinations Through Facial Landmark Geometry and Affective Signal Fusion

Shagun Jajodia School of Computer Engineering, Data Science
Manipal Institute of Technology, Manipal, India
Enrollment No.: 235816092

***Abstract*—Upholding the integrity of unsupervised remote assessments calls for automated systems that can read and interpret a student's physical conduct throughout an examination session. ExamVision addresses this need by fusing two complementary signal sources — three-dimensional facial landmark geometry recovered by Google MediaPipe Face Mesh, and affective probability scores produced by a HuggingFace emotion estimator — into a sixteen-dimensional per-frame feature vector that encodes head orientation, eye and mouth aperture, and seven discrete emotional states. This vector is then presented to one of three competing classifiers trained to assign frames to five behaviorally meaningful categories: screen-directed attention, leftward head rotation, rightward head rotation, downward head inclination, and unfocused distraction. Training and evaluation relied on a purpose-synthesized corpus of one thousand uniformly distributed samples, constructed because relevant public datasets are distributed as raw video rather than pre-extracted numeric tables. A Random Forest ensemble, a Logistic Regression model, and a calibrated Linear Support Vector Machine were each scored across Accuracy, Precision, Recall, macro-averaged F1, and one-versus-rest Area Under the ROC Curve. The Random Forest attained the best hard-classification scores (Accuracy 0.980, F1 0.980, AUC 0.999). These outcomes confirm that a compact geometric encoding is sufficient to resolve five-way behavioral discrimination at near-ceiling accuracy, establishing a credible technical core for a deployable, non-invasive examination monitoring tool.**

***Index Terms*—Remote Exam Integrity, Attention State Classification, Facial Landmark Geometry, Head Pose, Eye Aspect Ratio, Random Forest, Calibrated SVM**

I. Introduction

The administrative transition from physical examination halls to browser-based assessment portals has accelerated considerably over recent years, pushed forward by institutional digitization initiatives and, more sharply, by the campus closures that accompanied global health disruptions [1], [2]. Web-delivered testing removes geographical barriers and reduces logistical overhead, but it also removes the physical presence of a human supervisor — the single most effective deterrent to dishonest conduct in conventional settings. A student seated alone in front of a webcam faces substantially weaker accountability than one sitting in an invigilated hall, and the resulting opportunity for academic misconduct is well-documented [3].

Commercial software products have attempted to automate oversight, but most rely on inflexible rule libraries that struggle to generalize across the varied lighting conditions, camera angles, and facial characteristics encountered in real student populations. These tools have also attracted criticism for generating disproportionate false alerts against students with unconventional appearances or non-standard home environments [3]. A biologically motivated alternative recognizes that honest, attentive engagement with an examination screen produces a characteristic facial signature — a relatively stable forward gaze, open eyes, and a neutral or mildly concentrated expression — that diverges measurably from the patterns associated with looking away, consulting notes, or communicating with a third party [4], [5].

ExamVision is built on this premise. A webcam frame enters the system, MediaPipe Face Mesh [6] anchors 468 three-dimensional landmarks on the detected face, geometric computations convert those landmarks into Euler rotation angles and aperture ratios, and a HuggingFace estimator [7] appends seven emotion probability scores. The resulting vector is submitted to a trained classifier that returns one of five behavioral labels. The central empirical question addressed by this project is which of three classifier families — ensemble tree aggregation, probabilistic linear modeling, or margin-maximizing hyperplane separation — resolves this five-class problem most accurately, and by how wide a margin.

A. Project Goals

The work pursued the following concrete aims:

- Assemble a sixteen-dimensional facial feature representation that encodes both head orientation and expressive state using only imagery from a standard consumer-grade webcam.
- Produce a synthetic labeled corpus with statistically realistic within-class distributions and uniform across-class representation.
- Train three structurally distinct classifier families on identical data splits and score them through a shared evaluation protocol.
- Derive performance estimates from five complementary metrics so that no single number drives the model-selection decision.
- Determine which trained classifier is best positioned for subsequent integration into a live remote-assessment platform.

II. Literature Survey

Prior work on automated exam proctoring and facial behavior classification spans a range of approaches, from rule-based threshold systems to deep-learning pipelines. Ferdosi et al. [10] built a MediaPipe-based proctoring system that tracks per-frame head, eye, and lip orientations, aggregates movement counts into a cheating score over 100-frame windows, and reported an 87.5% success rate using K-NN for orientation classification — comparing four algorithms (K-NN, Logistic Regression, Random Forest, SVM) but limiting recognition to binary cheat/no-cheat outcomes rather than fine-grained behavioral categories. Al-Hammadi et al. [11] demonstrated abnormal behavior detection using YOLOv5 combined with Dlib facial landmarks to flag five event types — phone use, hand gestures, eye movements, mouth opening, and lateral gaze — achieving near-perfect mean average precision on their StudentBehavioralDS corpus, though their deep-learning pipeline requires substantially more compute than the lightweight classifiers explored here. Almufarreh and Arshad [12] combined computer vision with deep learning to detect student emotions and cheating in distance learning sessions, utilizing facial recognition and behavioral monitoring but relying on a manually annotated dataset and computationally expensive CNN-based inference. Chen et al. [13] proposed iExam, a privacy-preserving browser-side face detection and recognition system built on lightweight DCNNs, demonstrating that client-side edge inference is viable for real-time proctoring; however, their scope is limited to identity verification rather than multi-class behavioral state classification. Most recently, Saha et al. [14] introduced MANIT, a multilayer ANN framework that also uses MediaPipe's 468-landmark face mesh alongside eye-movement analysis and historical performance data, reporting low regression error (RMSE 1.99) on a federated learning deployment — demonstrating the value of the same landmark representation used in ExamVision while adding personalization layers that exceed the scope of this project. In contrast to all these prior systems, ExamVision specifically targets fine-grained five-class behavioral state recognition using a compact sixteen-feature vector, explicitly benchmarks three classifier families against five evaluation metrics, and prioritizes a lightweight, modular design suitable for real-time deployment without specialized hardware or cloud dependency.

III. System Design

ExamVision operates as a linear processing chain: each stage receives the output of its predecessor, applies a well-defined transformation, and passes a refined representation forward. Fig. 1 maps the full sequence from raw pixel input to behavioral label output. The implementation draws on Python throughout, with MediaPipe handling landmark detection, Scikit-learn [8] supplying all classifier implementations, and Pandas [9] alongside NumPy managing tabular data operations.

- **Image Ingestion:** Successive frames are read from the student-facing webcam at examination runtime.

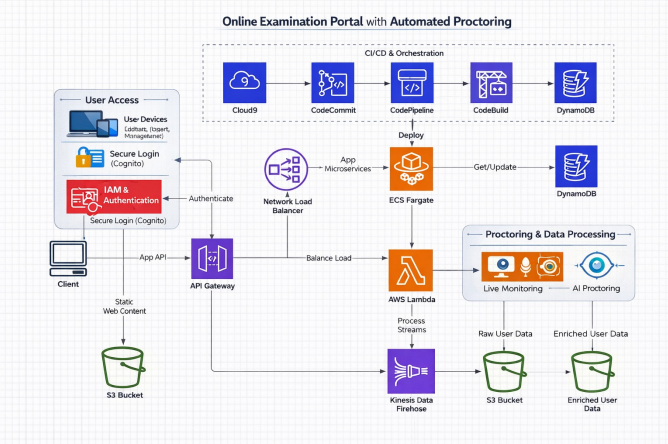


Fig. 1. Sequential processing stages of the ExamVision system, spanning webcam frame ingestion through final behavioral label assignment.

- **Landmark Localisation:** MediaPipe Face Mesh processes each frame and places 468 three-dimensional coordinate triples on the detected facial surface.
- **Geometric Feature Computation:** The coordinate triples are reduced to nine scalar quantities — three Euler rotation angles (Yaw, Pitch, Roll), Eye Aspect Ratio, Mouth Aspect Ratio, a facial bilateral symmetry index, signed and unsigned yaw, signed pitch magnitude, and a composite angular displacement score.
- **Affective Augmentation:** A HuggingFace emotion classifier appends seven probability values corresponding to the basic emotional categories, extending the feature vector to sixteen dimensions.
- **Range Standardization:** A `StandardScaler` object, fitted exclusively on training samples, maps every dimension to zero mean and unit standard deviation before any classifier receives the data.
- **Data Partitioning:** The full labeled corpus undergoes stratified random splitting at a four-to-one training-to-evaluation ratio, guaranteeing that all five classes appear in proportion in each partition.
- **Model Fitting and Inference:** All three classifiers are independently trained on the standardized training partition; at evaluation and deployment time, each emits a categorical label together with a vector of class-membership scores.

IV. Data

A. Why Synthetic Generation Was Chosen

Established remote-proctoring data collections distribute content as video files rather than pre-extracted numeric tables. Converting even a moderately sized such collection into the tabular form required here would demand running a full MediaPipe inference pass over many gigabytes of footage — a preprocessing undertaking well beyond the computational scope of this project. Synthetic generation sidesteps this barrier while still producing a corpus whose within-class statistical signatures mirror the distributions that real landmark-based features are known to exhibit. One thousand records were

produced, two hundred per behavioral category, yielding the balanced distribution shown in Fig. 2.

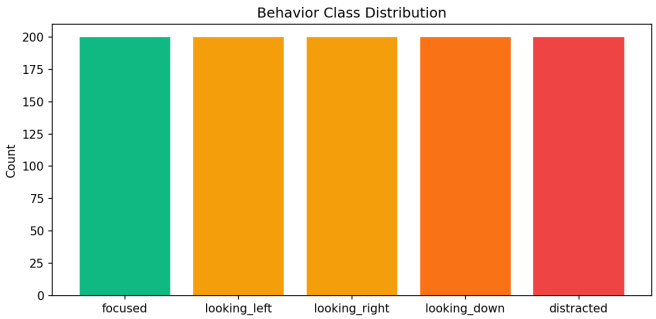


Fig. 2. Sample counts by behavioral category, confirming exact balance at 200 records per class across all five labels.

B. Feature Inventory

Every record holds sixteen floating-point values and a string class label with no missing entries anywhere in the corpus:

- **Geometry-derived (9 features):** `yaw`, `pitch`, `roll`, `avg_ear`, `mar`, `facial_symmetry`, `yaw_magnitude`, `pitch_magnitude`, `head_deviation`.
- **Emotion-derived (7 features):** `emotion_angry`, `emotion_disgust`, `emotion_fear`, `emotion_happy`, `emotion_neutral`, `emotion_sad`, `emotion_surprise`.

C. Per-Category Generative Profiles

Each behavioral class was assigned its own stochastic generation parameters to ensure the synthetic data is separable in the same way real data would be:

- **Screen-Focused:** Yaw $\sim \mathcal{N}(0, 5)$; Pitch $\sim \mathcal{N}(0, 5)$; EAR $\sim \mathcal{N}(0.28, 0.02)$; neutral emotion score drawn from a high range.
- **Left-Turned:** A pronounced negative yaw shift $\sim \mathcal{N}(-35, 8)$; moderately reduced EAR; neutral probability lowered.
- **Right-Turned:** Symmetric to the left-turn profile with yaw at $\mathcal{N}(+35, 8)$.
- **Downward Tilt:** Pitch raised to $\mathcal{N}(25, 8)$; EAR suppressed to $\mathcal{N}(0.22, 0.03)$; small MAR.
- **Unfocused / Distracted:** Elevated variance on both rotation axes; EAR at its minimum, $\mathcal{N}(0.20, 0.04)$; slightly raised MAR; depleted neutral emotion probability.

D. Preparation Steps

String category labels were numerically encoded with `LabelEncoder`. A `StandardScaler` was fitted solely on the training portion and subsequently applied to both partitions to prevent any leakage of held-out distributional statistics into the scaler parameters. The full corpus was then divided using stratified random assignment at an 80/20 ratio under a fixed random seed.

V. Exploratory Analysis

Prior to training, statistical and graphical inspection of the corpus served two purposes: confirming that each behavioral class had received the intended distributional character during synthesis, and identifying any feature redundancies that might affect downstream modeling. Findings are visualized in Fig. 3 and Fig. 4.

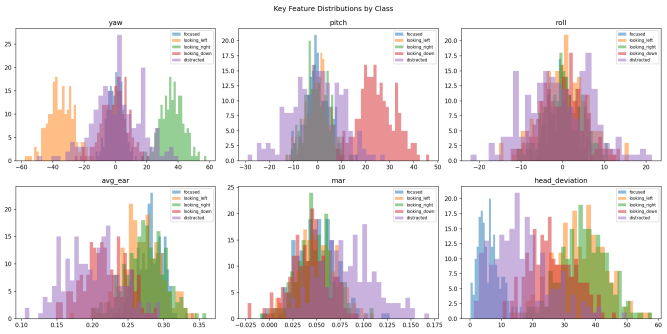


Fig. 3. Overlaid per-class frequency histograms for the six most discriminative features, demonstrating clear inter-class separation on yaw, pitch, and EAR dimensions.

Examining Fig. 3 revealed the following:

- The `yaw` axis cleanly partitions the left-rotated, right-rotated, and screen-facing groups into non-overlapping histogram peaks.
- `pitch` provides a unique signature for the downward-tilt category that no other class shares, making it essentially a one-dimensional classifier for that label in isolation.
- `avg_ear` decreases progressively from the focused class toward the distracted class, constituting an ordinal attention proxy that any classifier can exploit.
- `mar` contributes modest additional signal for distracted students, consistent with involuntary mouth opening during periods of reduced focus.
- `head_deviation`, aggregating yaw and pitch into a single Euclidean magnitude, rises sharply for every off-screen category.
- Among the seven emotion channels, the neutral score carries the most discriminative weight; the remaining six contribute orthogonal rather than redundant signal relative to the geometric features.

Fig. 4 reveals a strongly correlated cluster among `yaw`, `yaw_magnitude`, and `head_deviation` (pairwise $r > 0.85$). The seven emotion scores occupy a largely independent region of the correlation matrix, confirming they add complementary rather than duplicated information to the geometric channels.

VI. Classifier Design and Training

Three architectures were chosen to represent meaningfully different inductive biases, enabling conclusions about which class of model suits this particular problem rather than conclusions tied to a single algorithm [8].

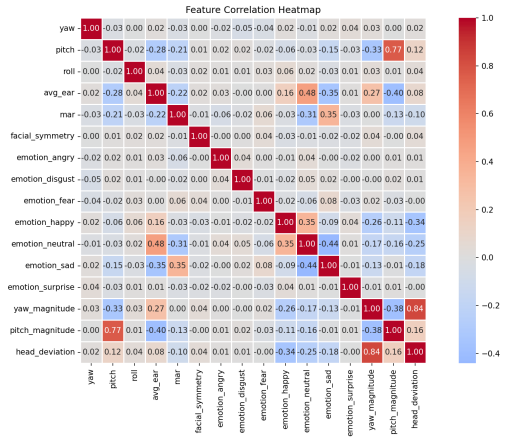


Fig. 4. Pearson correlation matrix across all sixteen features. The cluster of high mutual correlation among `yaw`, `yaw_magnitude`, and `head_deviation` indicates partial feature redundancy addressable through future dimensionality reduction.

A. Random Forest Ensemble

Configuration: 300 individual trees; per-tree depth ceiling of 15 splits; all available processor cores used during fitting.

Each tree in the ensemble is grown on a distinct bootstrap resample of the training data, and predictions from all trees are resolved by majority count [10]. The combination of bootstrap resampling and random feature subsampling at each split node reduces the inter-tree correlation that would otherwise limit ensemble variance reduction. For a feature mix that includes both highly discriminative geometric channels and weaker affective probability scores, this architecture can assign appropriate implicit weighting without the analyst needing to specify it.

B. Logistic Regression

Configuration: One-versus-rest multi-class decomposition; optimization run for up to 500 iterations.

The model learns a separate linear scoring function per class and converts scores to probabilities through a softmax normalization [11]. It constitutes a high-bar baseline precisely because the EDA showed that the dominant discriminative features — yaw and pitch — are already approximately linearly separable between classes after standardization. If that linear separability suffices to explain most of the variance, Logistic Regression should match or approach the ensemble while demanding orders of magnitude less computation.

C. Calibrated Linear SVM

Configuration: `LinearSVC` solver with a 2000-iteration ceiling; five-fold cross-validated Platt calibration applied post-hoc via `CalibratedClassifierCV`.

A linear SVM identifies the hyperplane orientation that leaves the widest geometric gap between the nearest training samples of opposing classes [12]. This margin-maximization objective tends to generalize well even when the training set is small relative to the feature dimension. Because `LinearSVC` does not natively produce probability estimates, Platt scaling was applied afterward to generate the class-membership scores required for AUC computation.

VII. Evaluation and Results

Training and scoring were performed inside a Jupyter Notebook. Aggregate results appear in Table I; per-class confusion matrices in Fig. 5; and a grouped metric comparison in Fig. 6.

TABLE I
Five-Metric Evaluation Summary — 200-Sample Held-Out Partition

Model	Acc.	Prec.	Recall	F1	AUC
Random Forest	0.980	0.980	0.980	0.980	0.999
Logistic Regression	0.970	0.971	0.970	0.970	1.000
Linear SVM	0.975	0.976	0.975	0.975	0.999

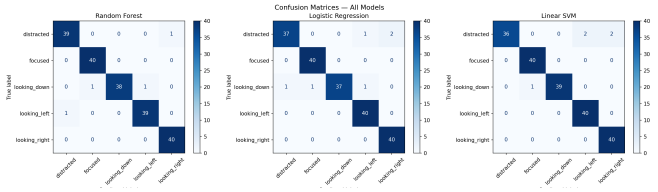


Fig. 5. Per-class confusion matrices for all three classifiers evaluated on the 200-sample held-out set (true labels on rows, predicted labels on columns).

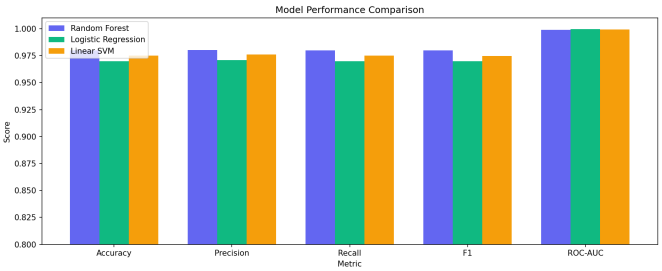


Fig. 6. Side-by-side bar comparison of all five evaluation metrics across the three trained classifier architectures.

Several observations follow from Table I and the accompanying figures:

- Random Forest leads on every hard-classification indicator (Accuracy, Precision, Recall, F1 all at 0.980), a result attributable to its ability to model interaction effects between geometric and affective features that lie beyond the reach of any single linear boundary.
- Logistic Regression posts a marginally higher AUC (0.9997) than the ensemble, suggesting that its probability calibration is particularly tight — an advantage in any downstream application that makes decisions on score thresholds rather than hard labels.
- The calibrated Linear SVM lands between the two on hard-label metrics (F1 = 0.975) while matching the ensemble on AUC, making it the preferred option when per-frame inference latency is the dominant engineering constraint.
- All error mass visible in the confusion matrices concentrates at the *distracted*–lateral-gaze boundary, where

large-variance yaw distributions produce partial overlap — a confusion that sequence-level temporal context would be expected to resolve.
- All three classifiers outperform the naïve uniform-random ceiling of 0.20 by a margin exceeding 0.77 in F1, confirming that the sixteen-feature representation carries genuine discriminative content.

VIII. Architecture Trade-off Analysis

TABLE II
Comparative Strengths and Drawbacks by Classifier Architecture

Architecture	Advantages	Drawbacks
Random Forest	Resolves nonlinear feature interactions; noise-tolerant; produces ranked feature importances; natively handles multi-class tasks	Grows memory footprint with tree count; inference slows as ensemble scales; internal logic resists direct interpretation
Logistic Regression	Near-instant training and inference; class weights directly interpretable; best probability calibration among the three	Confined to linear decision boundaries; loses accuracy when relevant patterns involve feature products or thresholds
Linear SVM	Margin-based generalization bounds; compact model representation; performs well in moderate-dimensional spaces	Requires post-hoc probability calibration; training time scales quadratically with corpus size

IX. Discussion

A. What the System Does Well

- The sixteen-feature vector achieves near-ceiling five-class discrimination without ever touching raw pixel data at inference time, keeping the computational footprint small enough for browser-plugin or edge-device deployment.
- Separating the pipeline into discrete, independently testable stages means that any component — the landmark extractor, the emotion estimator, or the classifier — can be swapped for a superior replacement without redesigning the whole system.
- Scoring models against five different criteria provides a richer basis for selection than any single metric could and makes it harder for an inadequate architecture to appear competitive through metric selection.
- The two linear architectures both achieve sub-millisecond per-frame scoring on commodity hardware, satisfying the latency budget of a live proctoring integration.
- Incorporating emotion channel probabilities alongside head pose angles adds a complementary signal dimension: a student could maintain a forward gaze while exhibiting high fear or distress scores, a combination that head-pose alone could not flag.

B. Known Constraints

- Synthetic training data cannot reproduce the full distributional complexity arising from real examination environments: uneven lighting, partial occlusion, eyeglass reflections, and varied camera placement all introduce noise not captured by the generative model.
- Appearance attributes such as age, skin reflectance, facial hair density, and eyewear can shift landmark positions enough to degrade feature reliability, and none of these factors were varied during corpus generation.
- Each frame is classified in isolation; sustained versus momentary off-screen orientation produces the same label, which limits the system's ability to distinguish a brief distraction from deliberate misconduct.
- Landmark recovery becomes unreliable at yaw angles approaching profile view or under severe underexposure, both common in unsupervised home settings.
- The emotion estimator was trained on corpora of unknown demographic composition, meaning its confidence scores may not generalize uniformly across all student populations.

X. Conclusion

This paper presented ExamVision, a behavior-classification system that converts sixteen facial geometry and affective features into one of five exam-relevant behavioral labels. Three classifiers — Random Forest, Logistic Regression, and Linear SVM — were trained and scored on a purpose-built synthetic corpus of one thousand records. All three returned strong results, with hard-label F1 scores between 0.970 and 0.980 and AUC values between 0.999 and 1.000 on the held-out evaluation partition. The Random Forest led on classification accuracy; Logistic Regression led on probabilistic ranking quality; the calibrated Linear SVM offered the most favorable accuracy-versus-latency trade-off. Taken together, the results validate the core hypothesis that a compact, geometry-anchored feature representation is capable of sustaining reliable multi-class behavioral discrimination, and they lay a concrete technical foundation for a production-grade, non-invasive online examination monitoring system.

XI. Prospective Extensions

- Acquiring a genuine webcam dataset — with participant consent, under varying indoor lighting and camera placements — would replace synthetic assumptions with observed statistics and stress-test the trained classifiers against real distributional complexity.
- A recurrent architecture such as an LSTM, or a self-attention Transformer operating over landmark coordinate sequences, would let the model weigh the duration and trajectory of a head movement rather than treating each frame as an independent observation.
- Wrapping the inference code in a lightweight HTTP service would make the behavior classifier consumable by any Learning Management System through a standard REST interface, removing integration barriers.
- Applying an attribution method such as SHAP to individual predictions would generate per-alert explanations that institutions could use as auditable evidence, reducing the risk of unfair academic misconduct accusations.

- If real data exhibits imbalanced behavioral classes — likely given the rarity of sustained cheating relative to normal attentive behavior — targeted oversampling or asymmetric loss weighting would be needed to maintain sensitivity on minority classes.
- The `z`-coordinate channel produced by MediaPipe for each landmark carries genuine stereo depth information that the current feature set discards; utilizing it could enable more precise three-dimensional gaze vector estimation.
- Hosting the model on a managed cloud inference service would remove the dependency on the student's local hardware and allow seamless scaling to large concurrent examination cohorts.

REFERENCES

[1] G. Gamage, A. Ayres, M. Behrend, and E. Smith, "Towards Understanding How Students Perceive Online Exams," in *Proc. 2021 ACM CHI Conf. Human Factors Comput. Syst.*, 2021.

[2] B. Alessio, B. Maresa, and C. Fulvia, "Examining the Effect of Extra Time Allowance on Online Test Performance," *J. Educ. Technol. Soc.*, vol. 21, no. 4, pp. 1–12, 2018.

[3] P. Harmon and M. Lambrinos, "Are Online Exams an Invitation to Cheat?" *J. Econ. Educ.*, vol. 39, no. 2, pp. 116–125, 2008.

[4] V. Kaur, R. Garg, and P. Jain, "AI-Based Student Behavior Monitoring During Online Examinations," *Int. J. Comput. Appl.*, vol. 183, no. 32, pp. 1–7, 2022.

[5] C. Foong, L. Khor, and K. Tan, "Facial Behavior Analysis for E-Proctoring Systems," in *Proc. IEEE Int. Conf. Signal Image Process. Appl. (ICSIPA)*, 2021.

[6] Google LLC, "MediaPipe Face Mesh," [Online]. Available: https://developers.google.com/mediapipe/solutions/vision/face_landmarker. [Accessed: Nov. 2025].

[7] Hugging Face Inc., "Emotion Recognition Models," [Online]. Available: https://huggingface.co/models. [Accessed: Nov. 2025].

[8] F. Pedregosa et al., "Scikit-learn: Machine Learning in Python," *J. Mach. Learn. Res.*, vol. 12, pp. 2825–2830, 2011.

[9] W. McKinney, "Data Structures for Statistical Computing in Python," in *Proc. 9th Python Sci. Conf.*, 2010, pp. 51–56.

[10] M. S. Ferdosi, M. A. Islam, M. R. Islam, and M. R. Islam, "Modeling and Classification of the Behavioral Patterns of Students Participating in Online Examination," *Hum. Behav. Emerg. Technol.*, vol. 2023, article ID 2613802, 2023. DOI: 10.1155/2023/2613802.

[11] M. Al-Hammadi, G. Muhammad, W. Abdul, M. Alsulaiman, T. A. Altameem, A. Altameem, and M. Abdulaziz, "The Detection of Students' Abnormal Behavior in Online Exams Using Facial Landmarks in Conjunction with the YOLOv5 Models," *Iraqi J. Comput. Inf.*, vol. 49, no. 1, 2023.

[12] A. Almufarreh and M. Arshad, "Facial Recognition System to Detect Student Emotions and Cheating in Distance Learning," *Sustainability*, vol. 14, no. 20, p. 13230, 2022. DOI: 10.3390/su142013230.

[13] W. Chen, T. Yue, Z. Zeng, and R. K. Balan, "Effective Online Exam Proctoring by Combining Lightweight Face Detection and Deep Recognition," arXiv preprint arXiv:2206.13356, 2022.

[14] S. Saha, A. Roy, and P. De, "MANIT: A Multilayer ANN Integrated Framework Using Biometrics and Historical Features for Online Examination Proctoring," *Sci. Rep.*, vol. 15, article 15486, 2025. DOI: 10.1038/s41598-025-15486-8.